

1. Pressure and the Ideal Gas Law

If you were to zoom in on a little piece of air, you would see a bunch of molecules, mostly nitrogen and oxygen, bouncing around randomly. The molecules would be spaced relatively far apart, occupying only about 0.1% of the volume of the air, and 99.9% empty space. The molecules regularly smack into each other and bounce off, with each individual molecule getting hit billions of times per second. When we zoom back out, somehow this messy, chaotic system turns into something quite orderly that can transport music from an instrument to our ears. We're going to try to understand at least a little bit about how this happens, and, eventually (but not in this note), how this messy microscopic picture actually can affect what we hear.

The first concept we want to work out is *pressure*. Pressure is defined to be the force per unit area a medium exerts on a surface. All sorts of things can cause pressures in physics, including light, magnetic fields, even quantum mechanics. Air and water pressures are the ones we most commonly experience directly, and in this class we'll just look at air pressure.

Example: Pressure on a Jar of Pickles

To get a sense of how pressure behaves, let's try to open a jar of pickles. The pressure at sea level is about 100,000 Pascals, or 100 kPa, where a Pascal is a Newton per square metre. This isn't really a very intuitive number, so let's break it down. Nobody I know actually thinks in Newtons; have you ever picked up a suitcase and said "this weighs about 200 Newtons"? Me neither. We can convert between mass and weight using Newton's second law:

$$F = ma \tag{1}$$

Here, F is the force, m is the mass, and a is the acceleration, the rate of change of velocity. Let's also be careful about units - m is in kg , a velocity is in m/s , so a *change* in velocity per unit time is $(m/s)/s$ or m/s^2 . In an equation, the units on both sides have to match, so the units of force must be the units of mass times the units of acceleration, or $kg - m/s^2$. This is a Newton - the force required to make a mass of 1 kg speed up by 1 m/s each second. The acceleration from Earth's gravity g at the surface of the Earth is about $9.8m/s$, which we'll round to $10m/s$. If we want the gravitational force, we can swap in g for a in Newton's second law, giving $F_g = mg$. If we have a weight of 1 kg then, it takes a force of about $10N$ to counter the force from gravity.

If we go back to atmospheric pressure, then $100,000N$ is the force required to hold up $100,000/10 = 10,000kg$ or 10 tons. That's a *huge* number, so why don't all the windows in your house blow out? Well, the pressure on the inside of your house is the same as the outside, so the forces from the pressures cancel out. If the pressure inside your house was lower, though, your windows would feel a big force and would most likely fail spectacularly.

The pressure inside a pickle jar is indeed much lower than atmospheric. If you've ever canned anything, you know that you heat up whatever you're canning, put it in a jar, seal it up, and let it close. When gasses cool, their pressure drops, so the pressure inside a jar is around 20-25% lower than atmospheric. Let's say the diameter of the jar lid is $10cm$ or $0.1m$, so the area is $\pi r^2 \sim 3(0.1/2)^2 \sim 0.008m^2$. The force is from the atmosphere is then $0.008 \times 10^5 = 800N$ downwards, or about an 80 kg weight on top of the lid. If the pressure inside is 75% of atmospheric, then there's an upward force of $600N$, so the total force on the lid is $200N$, equivalent to $20kg$.

When you unscrew the lid, you have to counter that force, which is why it's hard to open a jar the first time. Once you're open, though, the pressures are the same, and the next time you open is much easier!

2. Pressure from a Gas

Now that we (hopefully!) have at least a little bit of sense for what pressure feels like, let's look at where it comes from. If we zoom in on our pickle jar lid, we see a bunch of air molecules zipping around, bouncing off the lid. Every time a molecule bounces off the lid, its velocity changes¹, and a change in velocity means an acceleration. Newton's second law tells us that means the lid had to exert a force on the molecule. For a single molecule, the force is tiny, but there are a *huge* number of collisions each second, and their combined effects are not tiny. Let's try to work out what this looks like now.

We'll start by considering a long, thin tube (like a very, very long straw). The tube is closed at the ends, and the area of each end we'll call A . We've got a bunch of molecules bouncing around inside our tube, and we'll try to work out the force on the end. The first thing to notice is that we don't care about up/down or front/back motions if we're only looking at the end. Let's say we're looking at the left end - then any molecule near it moving to the left will bounce, and it will start moving to the right. Picture a bounce pass in basketball - the horizontal speed doesn't change, and the vertical speed flips from down to up.

Now that we know what we're looking for, let's try to work out the force. We're going to estimate a total mass bouncing off the end and the change in velocity, then plug those into Newton's second law to get a force. Let's call the "typical" speed in the x -direction v_x , the typical mass of each molecule m , and the number of molecules per unit volume n . We're going to make the simplifying assumption that half the molecules are moving to the left with v_x and half to the right. What happens at the left end cap? If we wait for a length of time t , then half of our molecules will move a distance $v_x t$ to the left, and half will move the same distance to the right. If we ignore collisions and assume each air molecule is just doing its own thing, then during our time interval t , every air molecule moving to the left within $v_x t$ of the end cap will bounce off the cap. Because the area of our tube is A , then every left-moving molecule with a volume of $v_x t A$ is going to bounce off. The number of molecules is going to be the number of things moving left per volume which is $n/2$ times the volume, so we have $v_x t A n/2$ collisions per second. The total mass that bounces off our end cap is the number of objects that hit the end times the mass per object, or $m v_x t A n/2$. The *change* in velocity is going to be $2v_x$, which happens over a time t , so the acceleration is $2v_x/t$. The force is now $m v_x t A n/2 \times 2v_x/t = m v_x^2 A n$. To get the force per unit area, we take the force and divide by the area A , leaving us with the simple expression:

$$P = m v_x^2 n \tag{2}$$

¹"But wait!" you say. "If I bounce off the lid, my speed is the same before and after I bounce!" Yes, that's true, but velocity is speed *and* direction. If I move left, bounce, then move right at the same speed, my velocity changed because my direction changed.

Now, you might remember that $E = 1/2mv^2$ is the kinetic energy of an object. We can rewrite the previous equation to be $P = 2(1/2mv_x^2)n = 2E_x n$ where E_x is the kinetic energy from the velocity in the x -direction. What is that energy, though? Well, you may remember being told that “temperature is a measure of the average energy”. Energy is in Joules, temperature (if you’re a physicist) is in Kelvin, and in our case we can go back and forth between them using

$$E_x = \frac{1}{2}k_B T \quad (3)$$

where T is temperature and k_B is one of the fundamental constants of nature, Boltzmann’s constant. The factor of $1/2$ comes from statistical mechanics, and its derivation is well beyond the scope of this class, but I hope you don’t mind taking the result on faith.

Once we have Equation 2, we plug in Equation 3, and we have the physicists’ version of the ideal gas law:

$$P = nkT \quad (4)$$

We can go halfway to a chemist’s version by noting if we have N things in a volume V , that $n = N/V$. Plug that into Equation 4 and we have

$$P = N/VkT \quad (5)$$

$$PV = NkT \quad (6)$$

In this case, N is the number of molecules we have in our volume V , but Chemists like to work with moles of things. I need Avogadro’s number $N_A = 6.02e23$ of things to get one mole. I can stick in two cancelling copies of N_A into the previous equation to get:

$$PV = N(N_A/N_A)kT \quad (7)$$

$$PV = (N/N_A)(kN_A)T \quad (8)$$

$$PV = N_m r T \quad (9)$$

where $r \equiv kN_A$ is known as the ideal gas constant, and N_m is the number of moles of stuff we have. Boltzmann’s constant is $1.38e-23J/K$, and so $r = 1.38e-23J/K \times 6.02e23/mole = 8.31J/K/mole$ – a number you may remember from chemistry class (if you took chemistry).

Before doing the recap, I’d like to make an aside, in case any of you are still worried about this derivation. Feel free to skip this, however. Pretending every molecule is moving with the same speed either in the $+x$ or $-x$ direction seems like a pretty big cheat. And it is! In reality, the molecules will all have random velocities, so to do this right, we’d need to average over those velocities. If you look at Equation 2, the right thing to do is to split up all our molecules into groups that are moving with the same velocity. We then sum over all those little velocity slices. Forces add, so pressures have to add, which leaves us with

$$P = \sum_{slice} m v_{x,slice}^2 n_{slice}$$

Because the mass is the same for everybody, we can pull it out of the sum. We can also define $f_{slice} = n_{slice}/n$, that is f_{slice} is the *fraction* of molecules with velocity v_{slice} . That turns our sum into:

$$P = mn \sum_{slice} v_{x,slice}^2 f_{slice} \quad (10)$$

That sum is exactly what we mean by the average of something, so in this case, what we should have used was the average value of v_x^2 , which we denote with angle brackets: $\langle v_x^2 \rangle$. That leaves us with

$$P = mn \langle v_x^2 \rangle = 2n \langle mv_x^2/2 \rangle \quad (11)$$

That expression is just the average kinetic energy from the x -velocity, which we know is $\frac{1}{2}kT$ (well, you would know if you had taken a course in statistical mechanics, but if you haven't, just take my word for it). When we made our seemingly horribly crude approximation that all molecules move with the same speed, we would generally expect that we'd get a slightly wrong (but close!) answer. In this case, though, once we saw what our answer was, we can actually see that we got exactly the right answer because the average we would need to do is exactly the average statistical mechanics already did for us.

We've now derived from first principles, only using one extra bit of information from statistical mechanics, that if I put a gas inside a container, the container will feel a pressure from the gas. The more stuff I put in the container, the higher the pressure will be. The hotter the gas, again, the higher the pressure will be. You'll notice that the mass of the things dropped out, so it doesn't matter *what* kind of gas I put in, just how many molecules.

One final note - you might say "Sure, this gives us the *average* force, but in reality it's a bunch of molecules hitting a wall. Does that matter?" It might! But probably not. Let's take the pickle jar again. The typical x -velocity is $\frac{1}{2}mv_x^2 = \frac{1}{2}kT$ which we can solve to get

$$v_x = \sqrt{\frac{kT}{m}} \quad (12)$$

The mass of a mole of protons/neutrons is around 1 gram². The average atomic weight of air is around 30 (nitrogen is 28, oxygen is 32), so the mass of an air molecule is $30/N_A \text{ g} = 30/N_A/1000 \text{ kg} \sim 5e-26 \text{ kg}$. We have to measure room temperature in Kelvin, so add 273 to temperature in Celsius to get $T \sim 293 \text{ K}$. That gives a typical speed in the x -direction of

$$|v_x| \sim \sqrt{\frac{1.38e-23 \times 293}{5e-26}} = 280 \text{ m/s} \quad (13)$$

If we count the collisions in one second, our pickle jar lid then picks up half the molecules in a volume of $280 \text{ m/s} \times 0.008 \text{ m}^2 \times 1 \text{ s} \sim 2.2 \text{ m}^3 = 2200 \text{ l}$. The density of air is about one gram per litre, so 2200 grams of air or 2.2 kg bounce off our pickle lid. We know that the mass of an air molecule is $5e-26 \text{ kg}$, so we have $2.2/5e-26 = 4.4e25$ air molecules bouncing off our pickle jar lid every second. Because the air molecules are moving randomly, some seconds we'll get a few more bouncing off of us, some seconds a few less. I won't derive it here, but the typical scatter in that goes like the square root of the number of things, so we'd expect $\sqrt{4.4e25} = 6.6e12$. While that may seem like a large number (6 trillion more air molecules hit me this second than hit me last second!), because the total number of collisions is so huge, you almost certainly would never notice it. What we can

²Originally, chemists wanted a mole of oxygen to be 16 grams, while physicists wanted a mole of hydrogen to be one gram. Much like Ottawa as the capitol of Canada was a compromise between Toronto and Montreal, physicists and chemists compromised and decided that a mole would be 12 grams of carbon-12.

measure is the force on the lid, and the *fraction* change in that is $6.6e12/4.4e25 = 1.5e-13$. That is, the randomness in the force on our jar lid is less than a part per trillion. That tiny change in force is so small you'd never notice it. However, if you zoom way in and look on very short times, the randomness *isn't* tiny, and you can see very small particles jitter around in what is called Brownian Motion. A young scientist worked out much of the math behind this in 1905 in order to predict the (random) motions of small objects buffeted by collisions with individual gas or water molecules. That scientist's name? Albert Einstein.